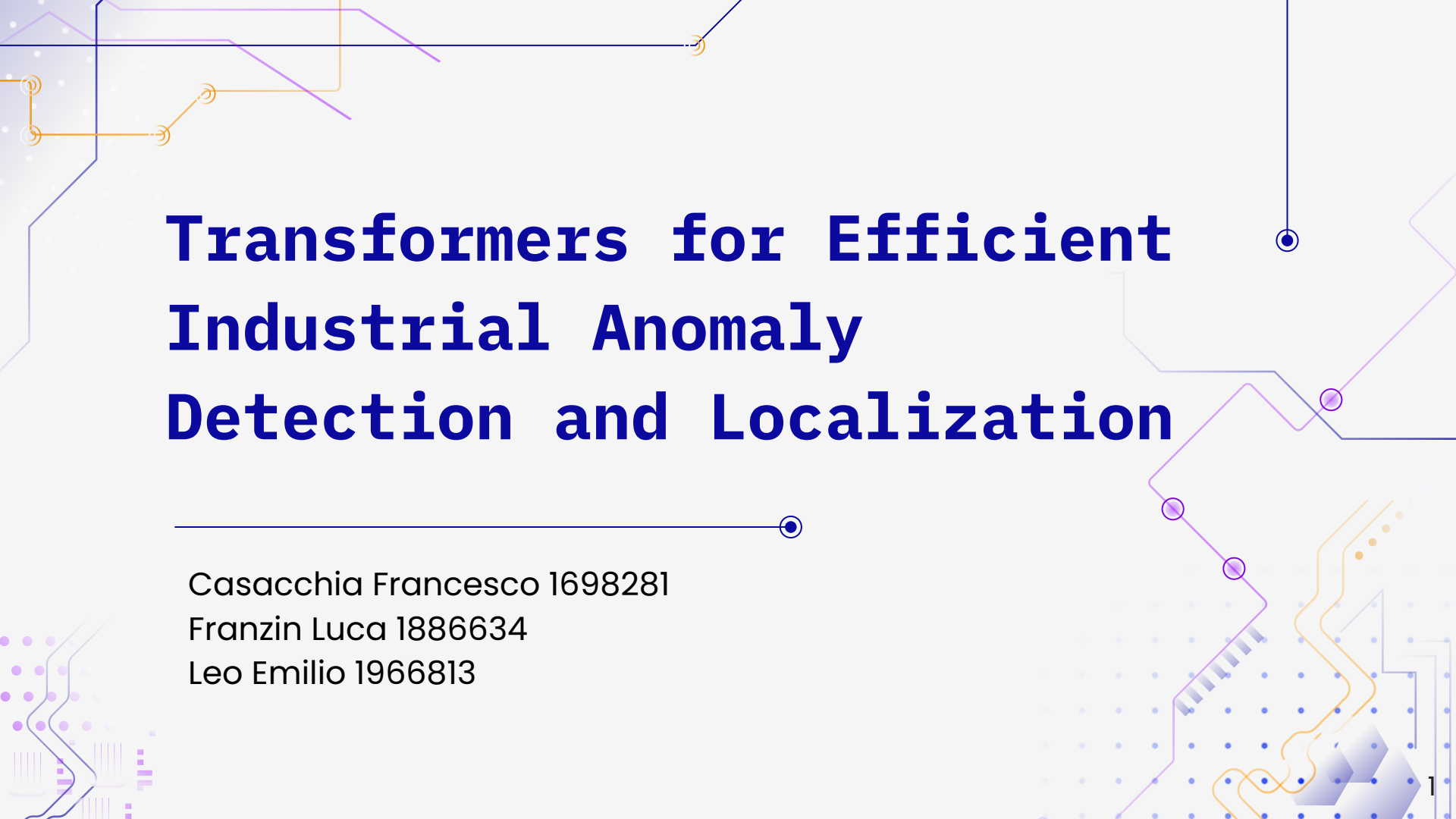




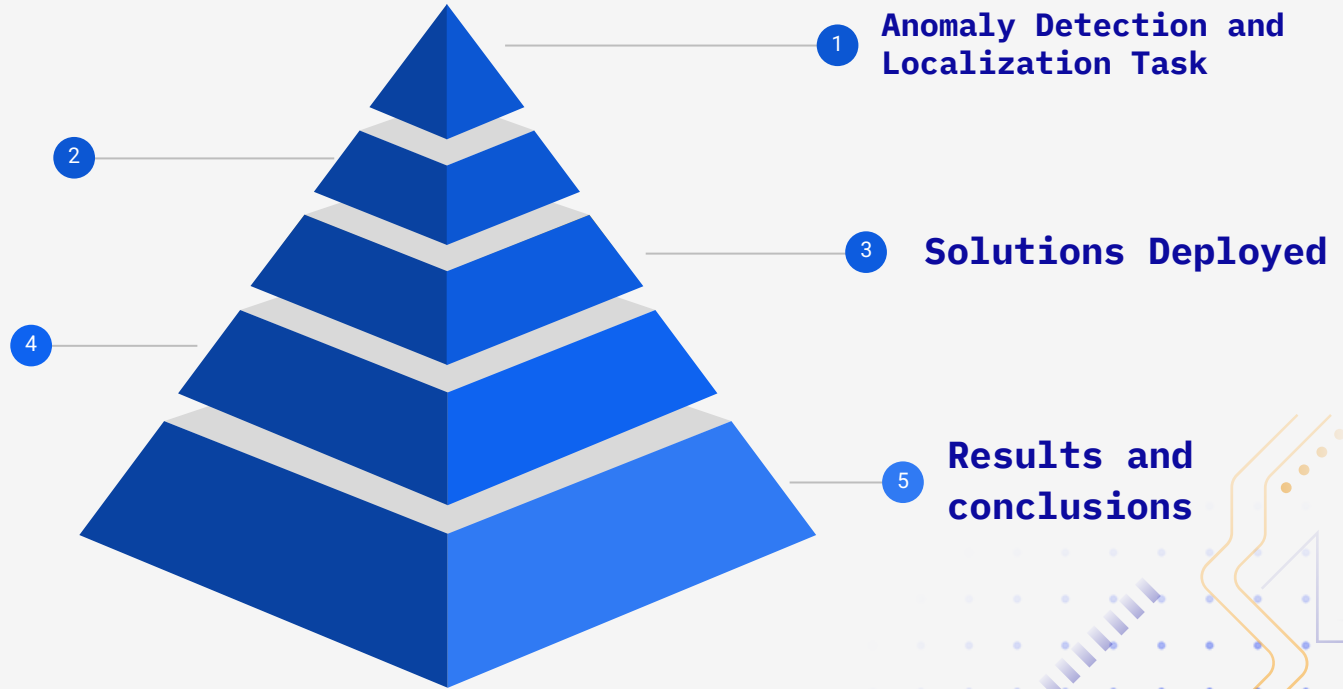
Transformers for Efficient Industrial Anomaly Detection and Localization

Casacchia Francesco 1698281

Franzin Luca 1886634

Leo Emilio 1966813

Presentation's organization



Description of
Datasets

Evaluation



01

Anomaly Detection and Localization Task

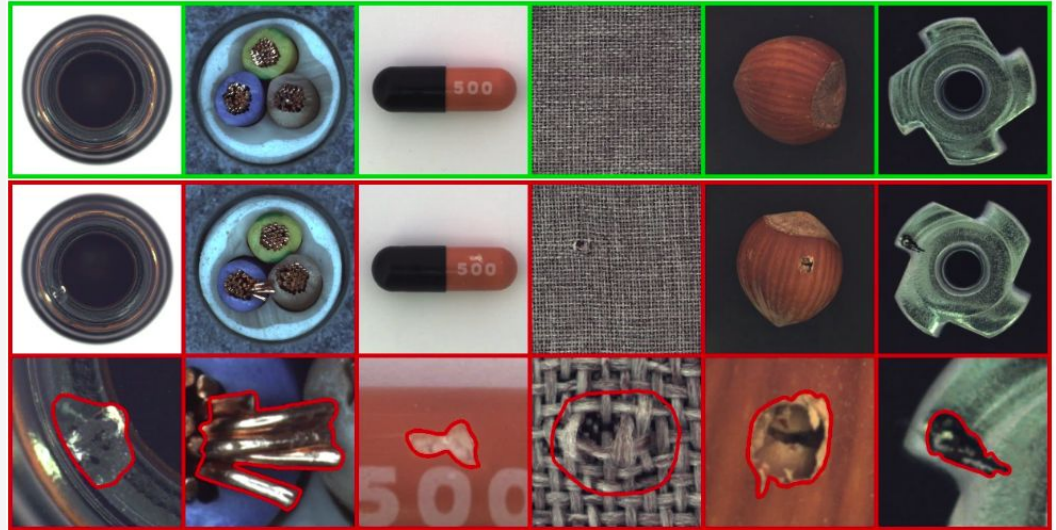


Problem of Anomalies

In the Industrial context,
each product can be built with
several errors having anomalies:

But there is no way to know
precisely what kind of anomaly
can occur

For this reason we need to find
models that only “know” what
“normality” looks like



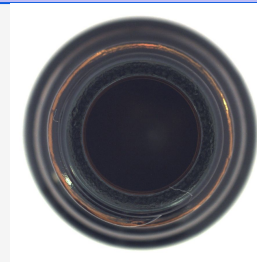
Detection and localization anomalies

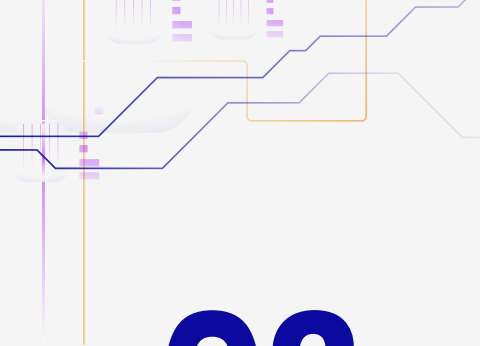
Detection and Localization tasks

Localizing pixels which
contains anomalies



classifying anomalous
images from normal
images



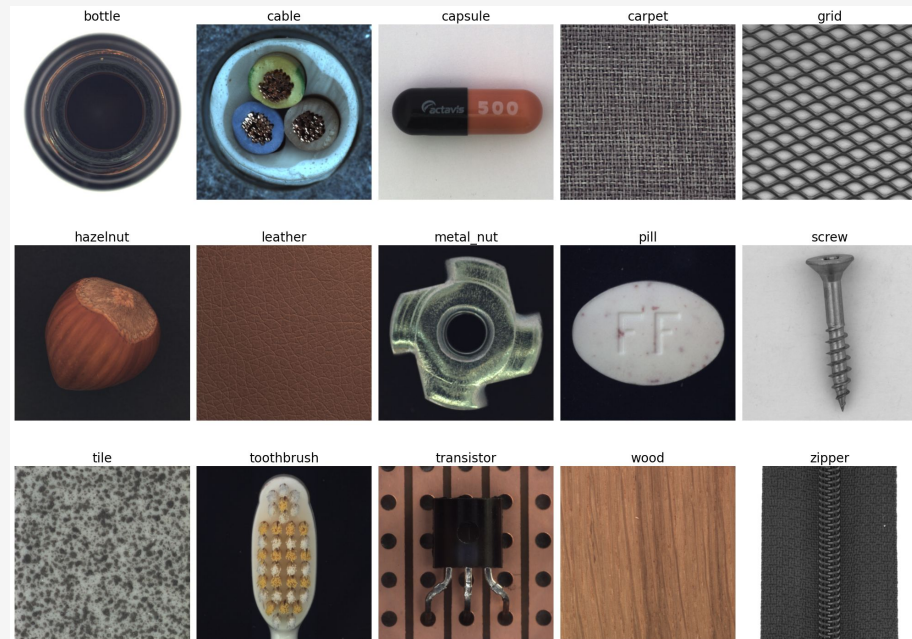
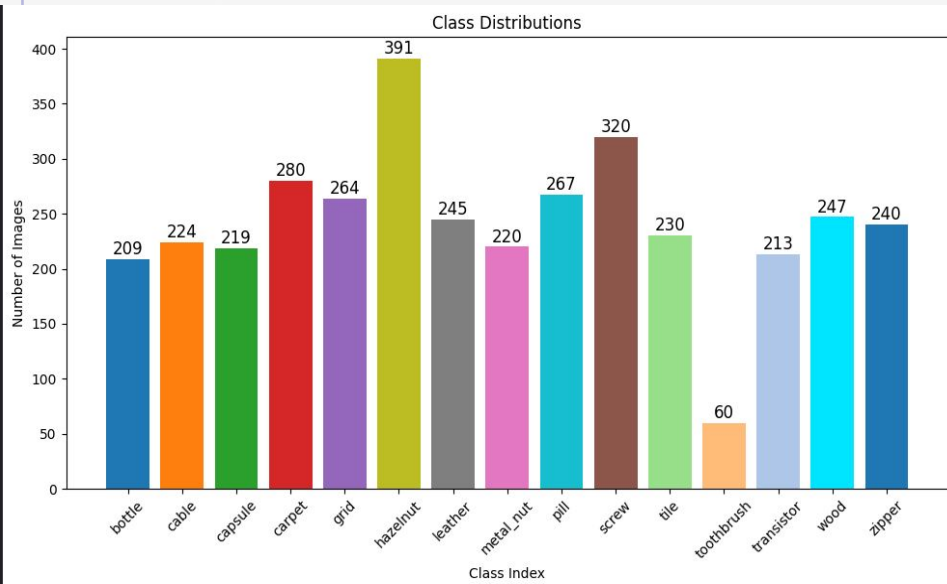


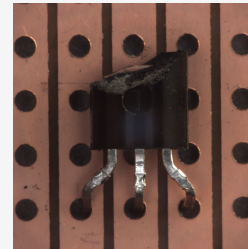
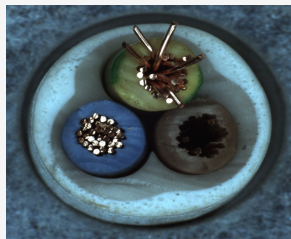
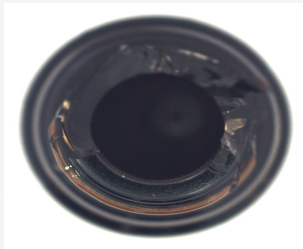
02

Description of Dataset

MVTec-AD dataset

- MVTec is characterized by 15 categories (divided in 10 object and 5 texture classes) of industrial products. In total it contains 3629 normal images and 1725 abnormal images.

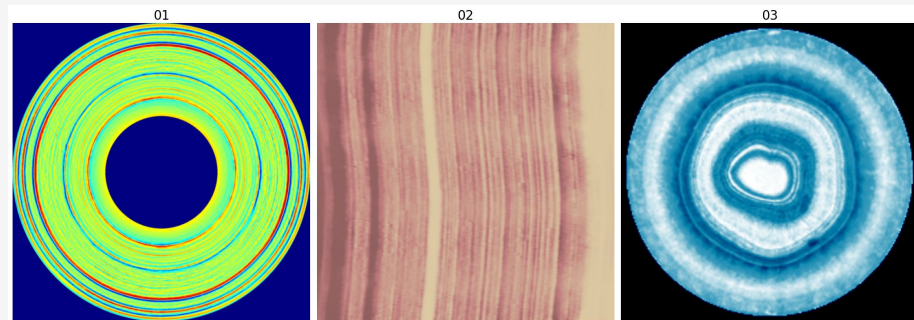
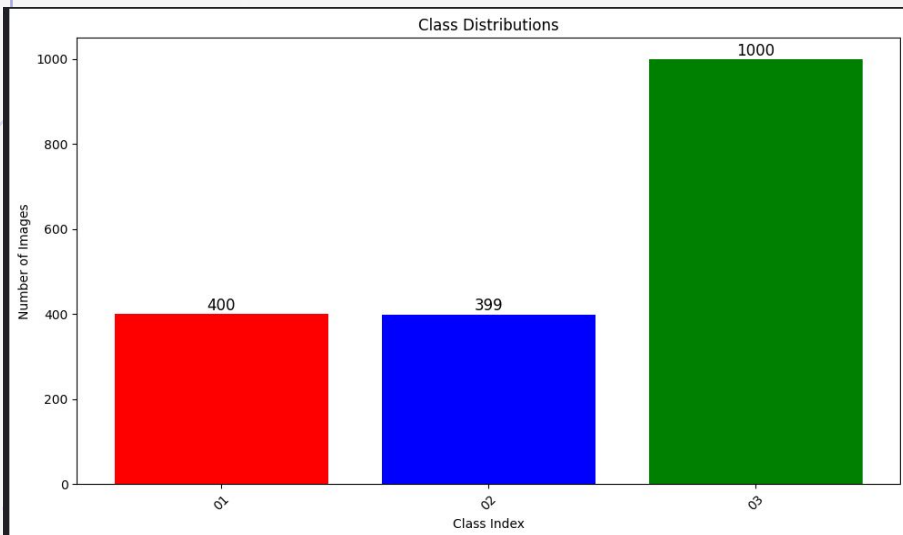


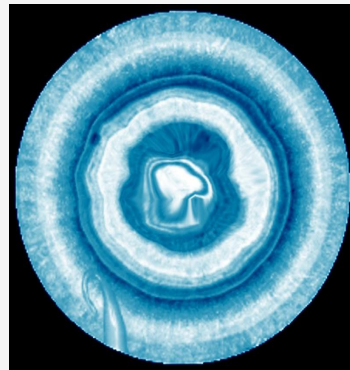
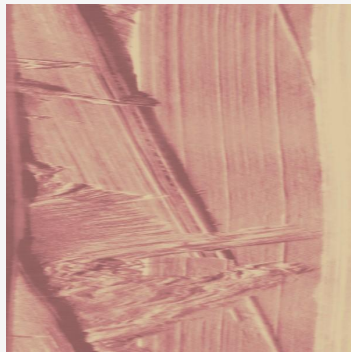
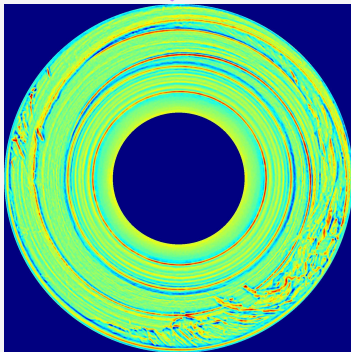


**These are some examples of some
anomalies in the MVTec-AD database**

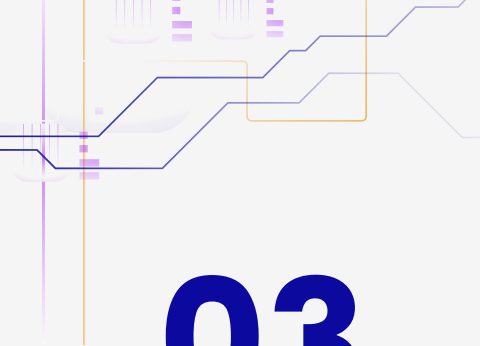
BTAD dataset

- BTAD contains 3 industrial products where product 1,2,3 have 400, 1000, 399 train images





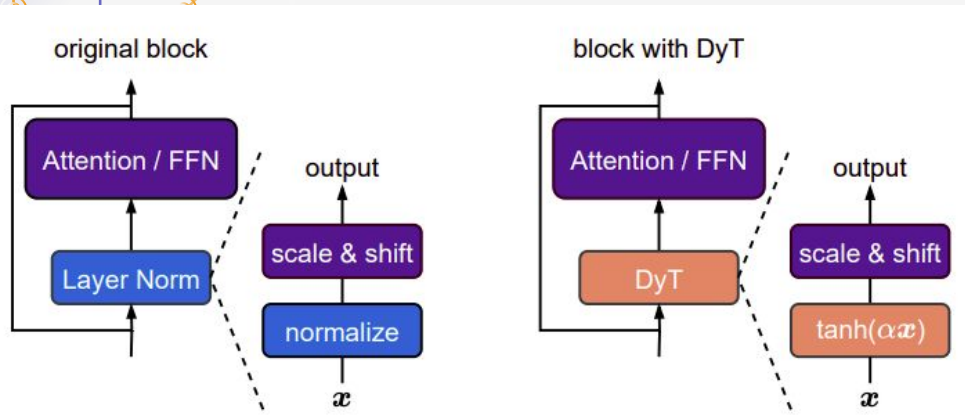
**These are some examples of some
anomalies in the BTAD database**



03

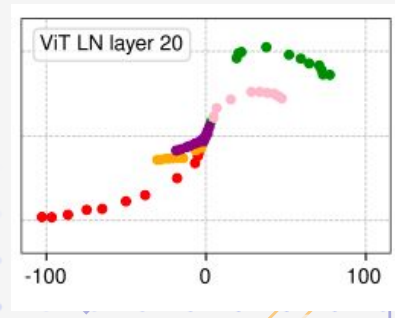
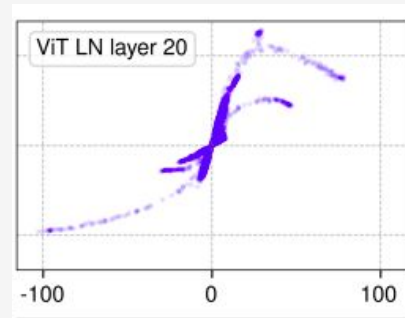
Solutions Deployed

Regarding the use of the DyT



It is defined as follows:
$$\text{DyT}(x) = \gamma * \tanh(\alpha * x) + \beta$$

From the similarity the shapes of distribution values of normalization layers in Vit we can substitute it with Dynamic Tanh(DyT) function.



Overview of proposed models

1

AnoVit:
Vit transformer +
Decoder

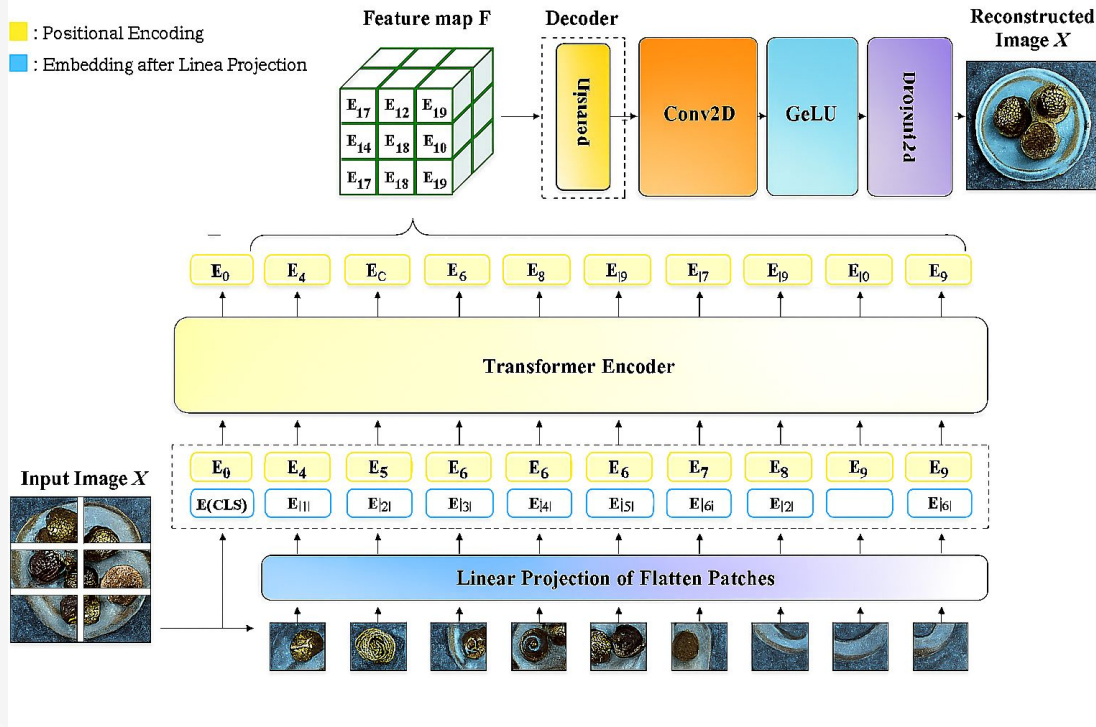
2

ADTR-Heavy:
Pretrained backbone
CNN + Variant Vit
transformer
(Encoder-Decoder)

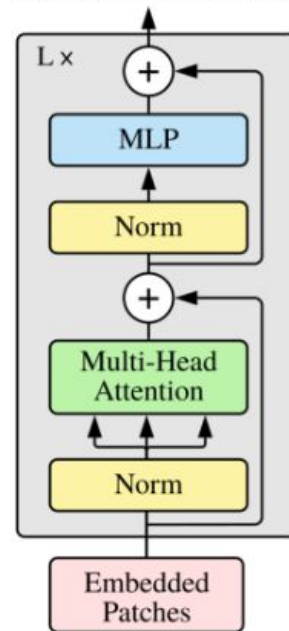
3

ADTR-BiFPN:
pretrained backbone
CNN + BIFPN layer+
Variant Vit
transformer(Encoder-D
ecoder)

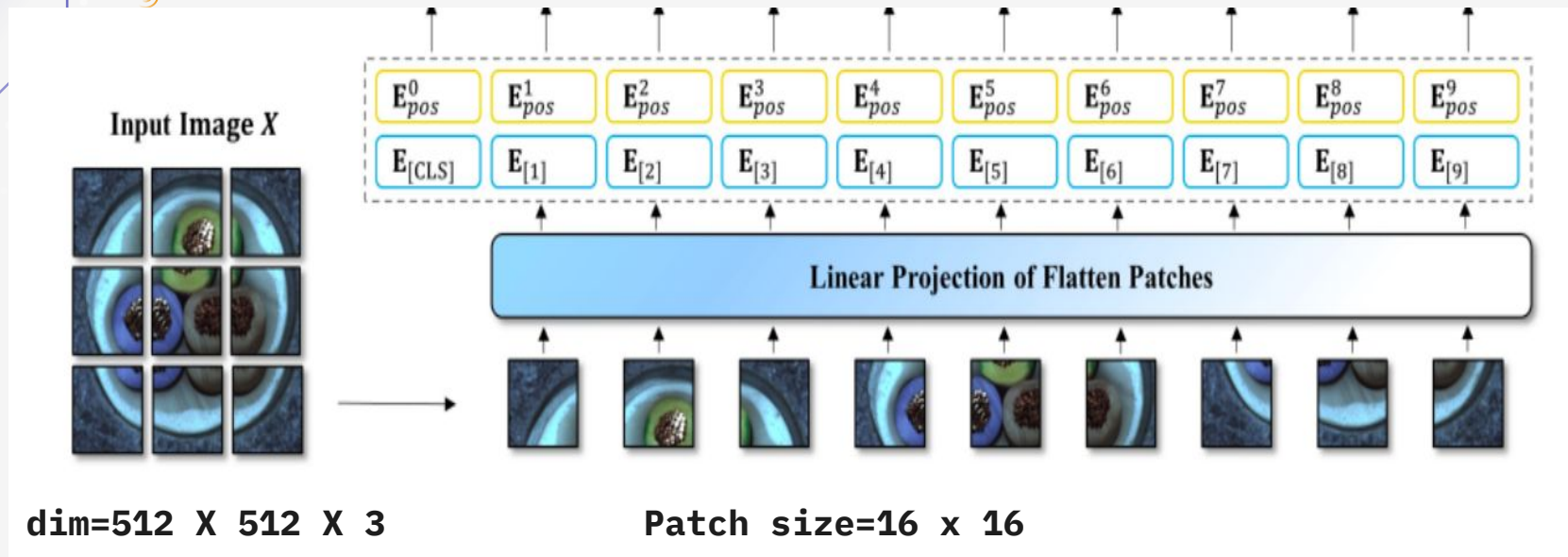
Anovit



Transformer Encoder



ANOVit: Embedding+Positional Encoding

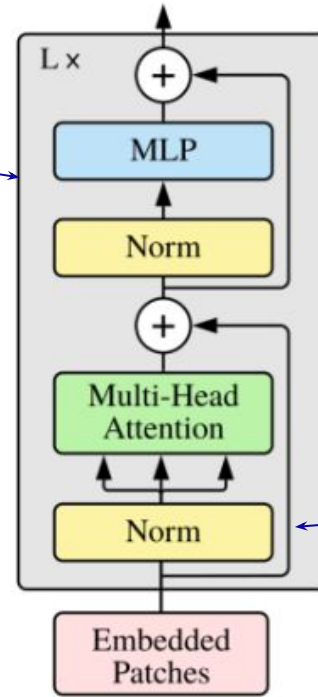


We have applied a convolution 2D to project each patch into d_{model} transformer obtaining flattened vector of dim: $d_{model} \times P$

ANOVIt: Transformer Encoder

MLP architecture is defined with one input layer, one hidden layer with $(d_{\text{model}} \times 4)$ and one output layer of dimension d_{model} . We have used GeLU activation function

Transformer Encoder

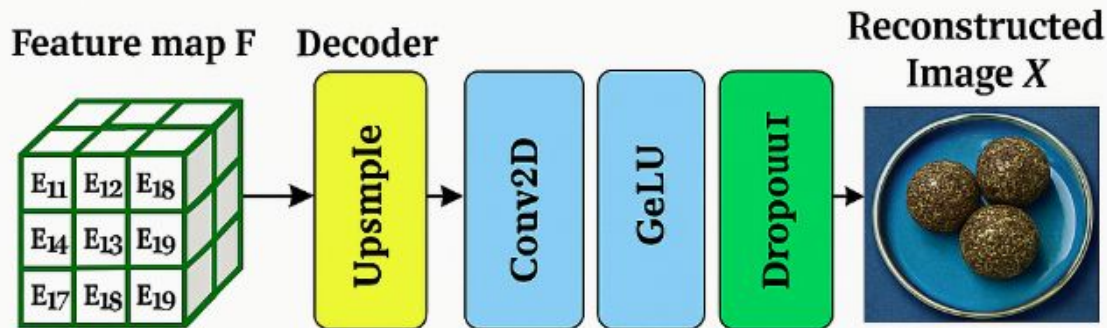


We have used Layer Normalization or DyT approach

Dim embedded patches= $(P \times d_{\text{model}})$

ANOVit: Decoder

Dim input:
 $d_{\text{model}} \times 16 \times 16$



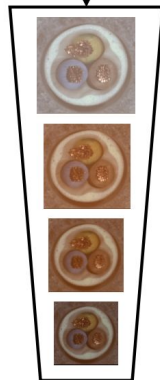
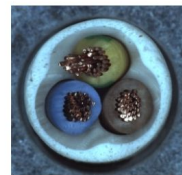
Dim Output
Image:
 $512 \times 512 \times 3$

From output of
encoder, we will
remove CLS token

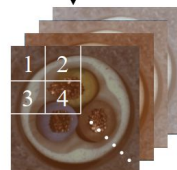
We create 4
upsampling block to
reach input image
dimension

ATDR-Heavy

**B5 Efficient
as a backbone**



Resize &
Concat



Multi-scale
Feature Map

Reshape

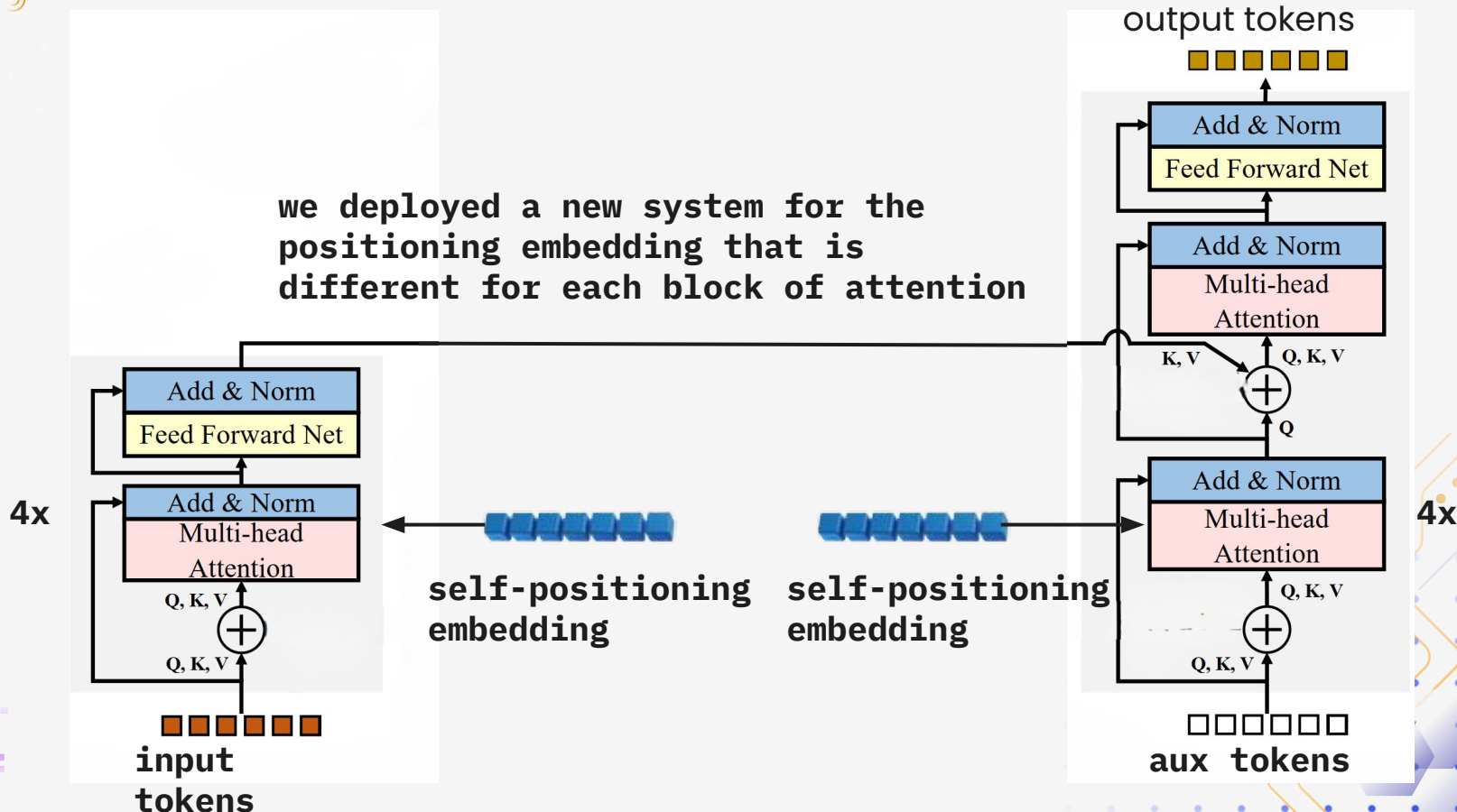


**patches of
24x24x816**

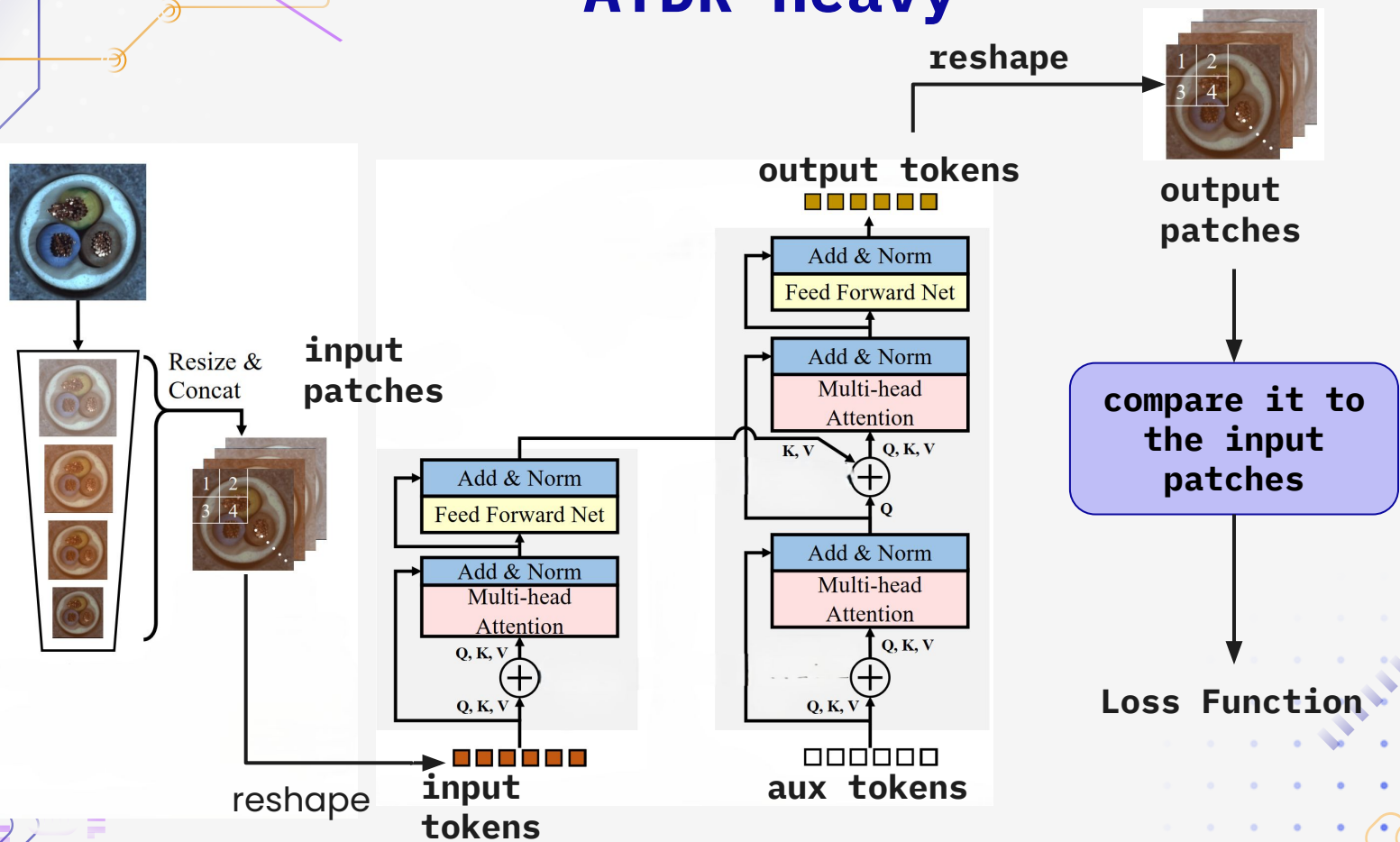
**576 tokens
each with
dimension 1200**

ATDR-Heavy

we deployed a new system for the
positioning embedding that is
different for each block of attention

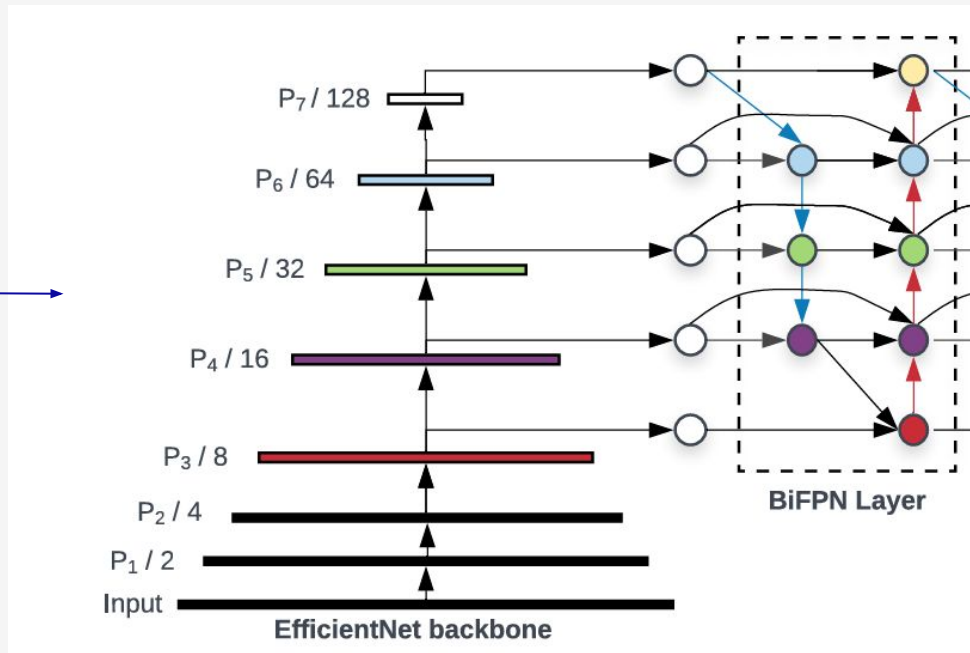


ATDR-Heavy



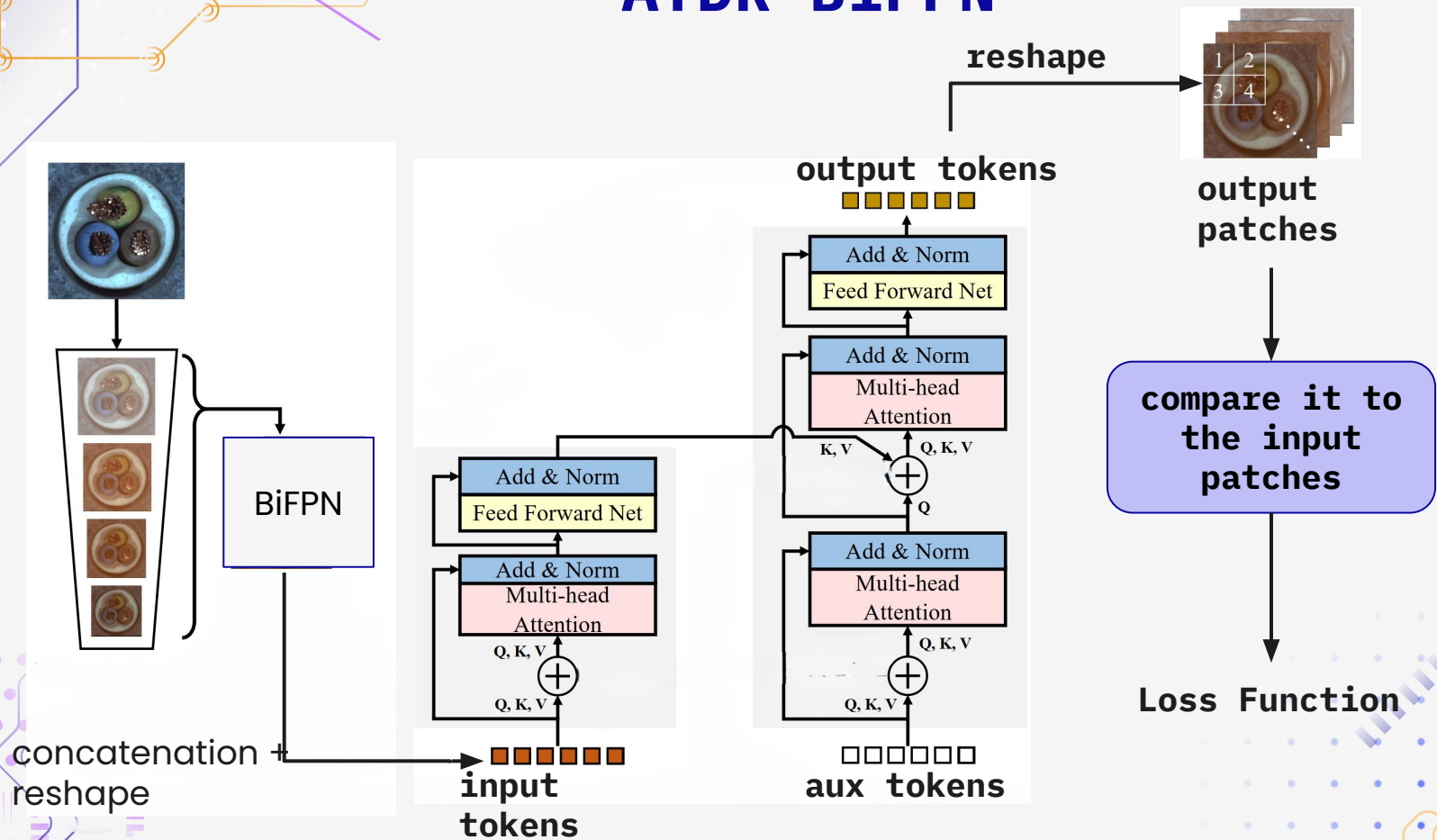
ATDR-BiFPN

**B5 Efficient
as a backbone**



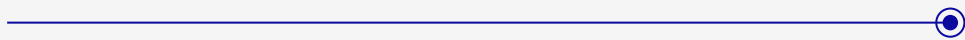
**single BiFPN
with Weighted
Feature Fusion
module**

ATDR-BiFPN

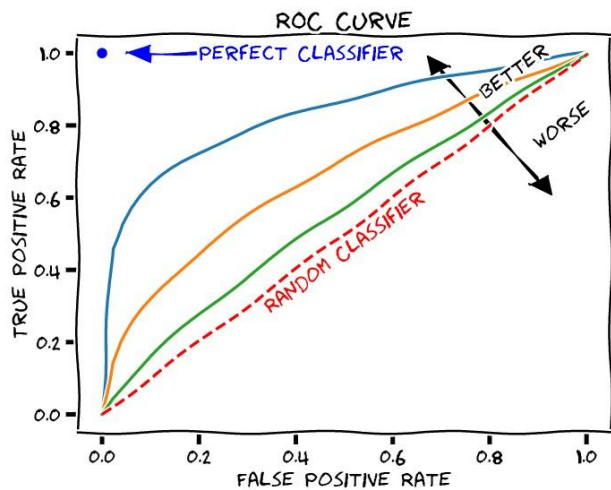


04

Evaluation



Pixel AUROC and Image AUROC



in order to compute the performances of anomaly detection two techniques are usually used: pixel auroc and Image Auroc.

Image AUROC is the area under the ROC curve calculated using the True positive and False Positive given an anomaly score

Pixel AUROC is the area under the ROC curve calculated using the True positive and the False positive of each pixel given an anomaly map

Anomaly Map and Anomaly Score (ADTR)

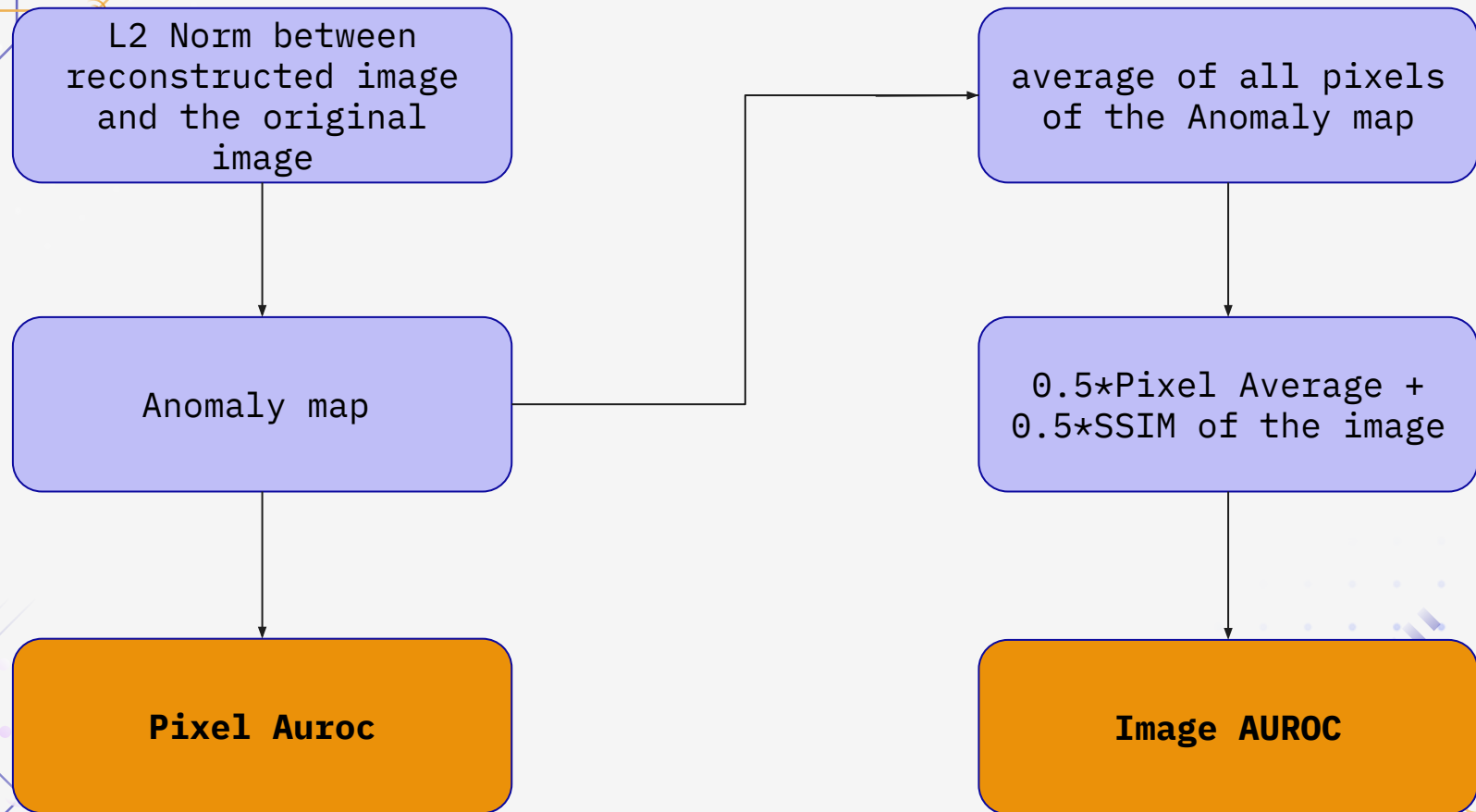
we take the **L2 norm** between the output features and the input features to obtain a 24x24 anomaly map

that we apply some post processing: **gaussian blur** and **bilinear interpolation** to upsample it to 416x416 dimension

we take the max value among all pixels as our **anomaly score**

we take the 416x416 tensor as our **anomaly map**

Anomaly Map and Anomaly Score (AnoVit)



05

Results and conclusions

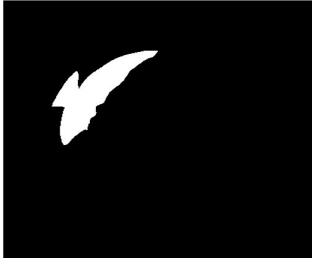
Some of our Results

Sample 5: (Anomaly)

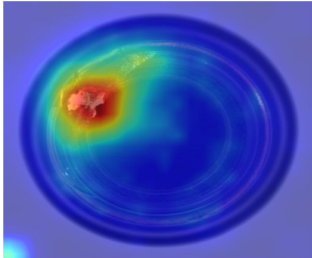
Original Image



Ground Truth

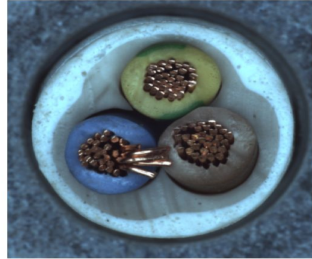


Predicted Anomaly

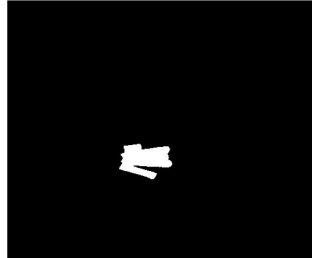


Sample 2: (Anomaly)

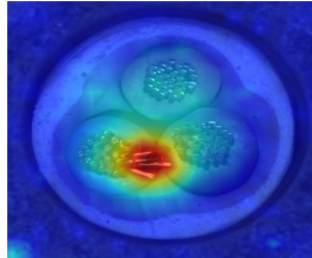
Original Image



Ground Truth



Predicted Anomaly

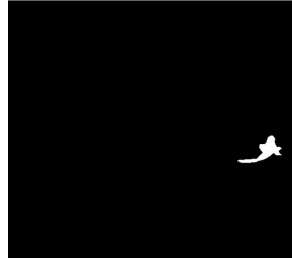


Sample 8: (Anomaly)

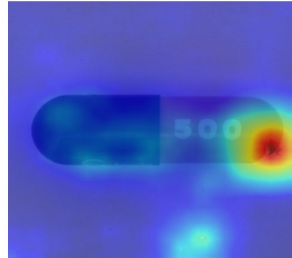
Original Image



Ground Truth



Predicted Anomaly

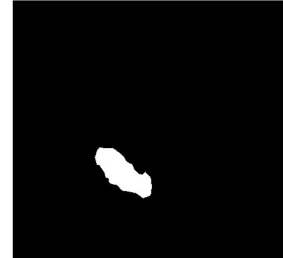


Sample 2: (Anomaly)

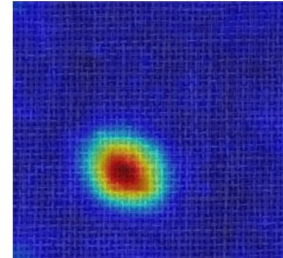
Original Image



Ground Truth



Predicted Anomaly

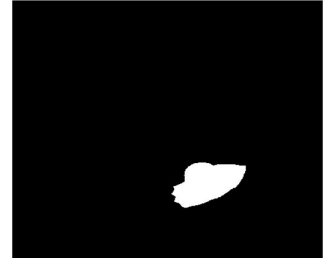


Sample 5: (Anomaly)

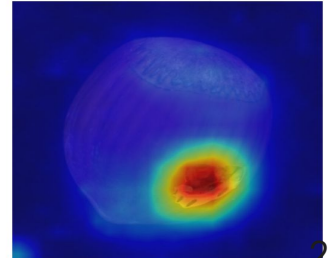
Original Image



Ground Truth



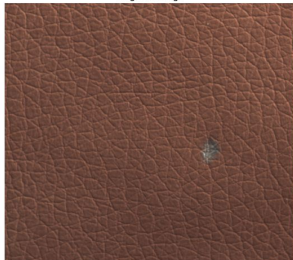
Predicted Anomaly



Some other Results

Sample 1: (Anomaly)

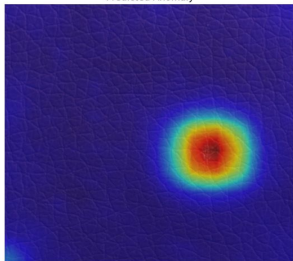
Original Image



Ground Truth

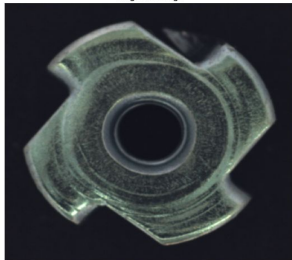


Predicted Anomaly

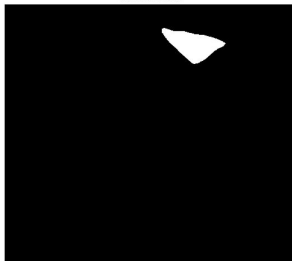


Sample 5: (Anomaly)

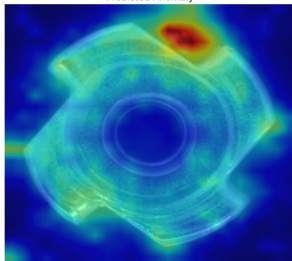
Original Image



Ground Truth



Predicted Anomaly

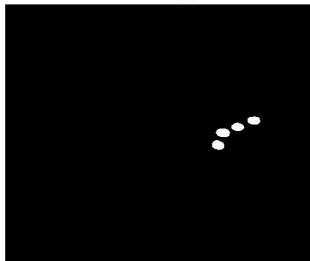


Sample 2: (Anomaly)

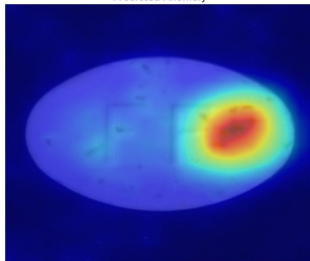
Original Image



Ground Truth



Predicted Anomaly



Sample 6: (Anomaly)

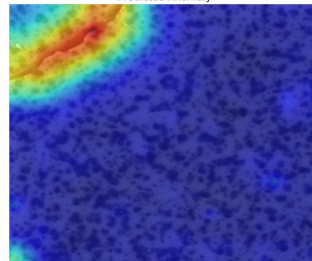
Original Image



Ground Truth

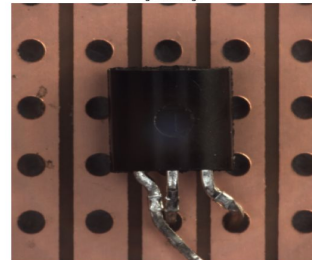


Predicted Anomaly

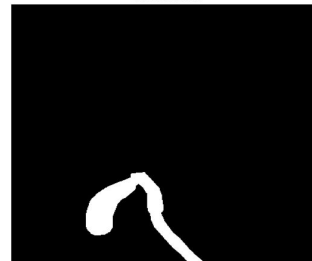


Sample 7: (Anomaly)

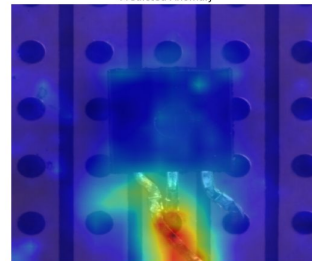
Original Image



Ground Truth

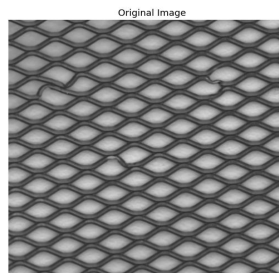


Predicted Anomaly

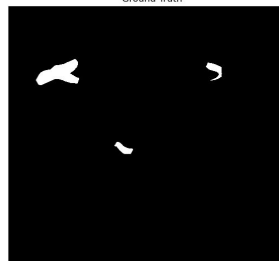


and some Anovit Results

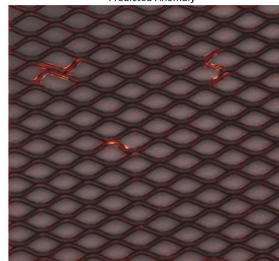
Sample 8: (Anomaly)



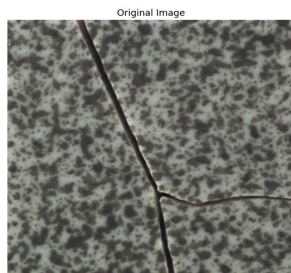
Ground Truth



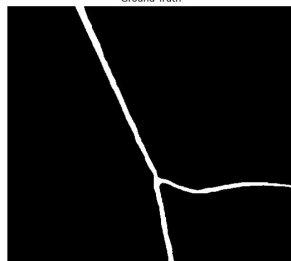
Predicted Anomaly



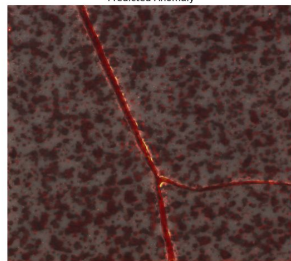
Sample 1: (Anomaly)



Ground Truth



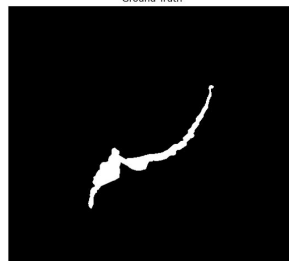
Predicted Anomaly



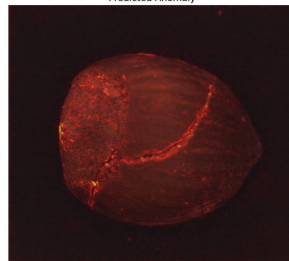
Sample 4: (Anomaly)



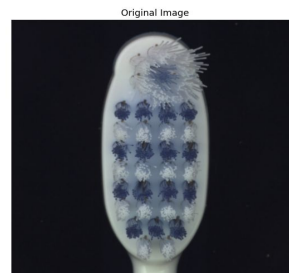
Ground Truth



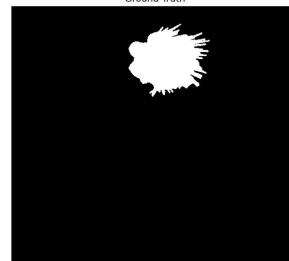
Predicted Anomaly



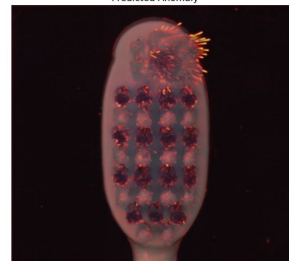
Sample 6: (Anomaly)



Ground Truth



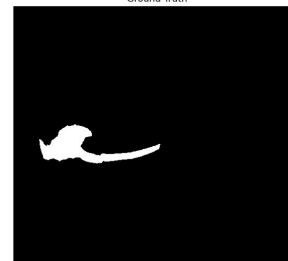
Predicted Anomaly



Sample 2: (Anomaly)



Ground Truth



Predicted Anomaly



Image AUROC MVTec-AD

model	bottle	cable	capsule	carpet	grid	hazelnut	leather	metal nut	pill	screw	tile	toothbrush	transistor	wood	zipper	average
AnoViT	0.83	0.74	0.73	0.5	0.52	0.88	0.85	0.86	0.72	1	0.89	0.74	0.83	0.95	0.73	0.78
AnoViT_our	0.972	0.686	0.7068	0.545	0.766	0.824	0.5007	0.925	0.7119	0.978	0.912	0.9278	0.895	0.894	0.894	0.800
AnoViT_our_Dyt	0.983	0.696	0.693	0.4141	0.782	0.729	0.509	0.819	0.7155	1	0.945	0.8694	0.896	0.904	0.899	0.801
ADTR	1	0.925	0.925	1	0.978	0.999	1	0.945	0.933	0.942	1	0.939	0.98	0.999	0.97	0.969
ADTR_Heavy	1	0.9925	0.9465	0.9932	0.8897	0.9996	1	0.9976	0.9476	0.9365	0.996	0.9833	0.9733	0.9868	0.9856	0.975
ADTR_Heavy_Dyt	1	0.9918	0.927	0.9948	0.8981	0.9996	1	0.9966	0.9493	0.933	0.9964	0.9889	0.9796	0.986	0.9835	0.975
ADTR_BiFPN	1	0.9754	0.9828	0.998	0.8404	1	0.9983	1	0.9667	0.951	0.9892	0.9944	0.9617	0.9912	0.9824	0.975
ADTR_BiFPN_Dyt	1	0.9766	0.9805	0.9988	0.8471	1	0.9983	1	0.9678	0.9529	0.9899	0.9944	0.96	0.9912	0.9798	0.976
DREAM	0.992	0.918	0.985	0.97	0.999	1	1	0.987	0.989	0.939	0.996	1	0.931	0.991	1	0.98
RD4AD	1	0.95	0.963	0.989	1	0.999	1	1	0.966	0.97	0.993	0.995	0.967	0.992	0.985	0.985

Pixel AUROC MVTec-AD

model	bottle	cable	capsule	carpet	grid	hazelnut	leather	metal nut	pill	screw	tile	toothbrush	transistor	wood	zipper	average
AnoViT	0.86	0.89	0.91	0.65	0.83	0.94	0.898	0.88	0.86	0.92	0.57	0.90	0.80	0.85	0.769	0.83
AnoViT_our	0.810	0.6636	0.808	0.62	0.785	0.885	0.802	0.819	0.8359	0.807	0.723	0.895	0.626	0.712	0.729	0.768
AnoViT_our_Dyt	0.762	0.666	0.843	0.629	0.819	0.917	0.813	0.821	0.798	0.768	0.753	0.889	0.684	0.706	0.726	0.772
ADTR	0.985	0.941	0.937	0.988	0.981	0.993	0.993	0.978	0.963	0.932	0.97	0.961	0.94	0.973	0.973	0.963
ADTR_Heavy	0.9634	0.9687	0.985	0.9817	0.9368	0.9843	0.9754	0.972	0.9851	0.983	0.9302	0.9865	0.9803	0.9135	0.965	0.967
ADTR_Heavy_Dyt	0.9653	0.9717	0.9849	0.9828	0.942	0.9846	0.9761	0.9727	0.9858	0.9831	0.9323	0.9865	0.9836	0.9148	0.9677	0.969
ADTR_BiFPN	0.9622	0.9566	0.9876	0.9795	0.9583	0.9837	0.9735	0.9681	0.9868	0.9852	0.9282	0.9903	0.9044	0.9166	0.9752	0.964
ADTR_BiFPN_Dyt	0.9641	0.9574	0.9881	0.9794	0.9582	0.9836	0.9735	0.969	0.9871	0.9853	0.9279	0.9908	0.9214	0.9169	0.9746	0.965
DREAM	0.997	0.947	0.978	0.985	0.987	0.996	0.996	0.979	0.975	0.891	0.981	0.989	0.94	0.968	0.992	0.974
RD4AD	0.992	0.925	0.947	0.972	0.984	0.989	0.991	0.987	0.966	0.927	0.967	0.971	0.95	0.981	0.979	0.966

Image AUROC BTAD

Model	01	02	03	Average
ANOVit	0.9310	0.7868	0.9992	0.900
ANOVit_DyT	0.9495	0.7802	0.9990	0.901
ADTR_Heavy	1.0	0.8867	0.9943	0.96
ADTR_Heavy_DyT	1.0	0.8850	0.9939	0.96
ADTR_BiFPN	0.9990	0.9137	0.9968	0.969
ADTR_BiFPN_DyT	0.9990	0.9113	0.9968	0.969

Pixel AUROC BTAD

Model	01	02	03	average
ANOVit	0.8656	0.7347	0.7682	0.780
ANOVit_DyT	0.8608	0.7554	0.7223	0.780
ADTR_Heavy	0.9525	0.9574	0.9962	0.968
ADTR_Heavy_DyT	0.9551	0.9585	0.9962	0.969
ADTR_BiFPN	0.9575	0.9623	0.9960	0.971
ADTR_BiFPN_DyT	0.9594	0.9622	0.9961	0.972

Conclusions

With all three models we managed to achieve the performances of the original paper or surpass it in some cases

Dyt managed to **slightly improve** some classes overhaul but on average the difference is about **0.2%**

but no noticeable improvement regarding inference or training time



06

References



References

Goldblum, M., Sourì, H., Ni, R., Shu, M., Prabhu, V., Somepalli, G., Chattopadhyay, P., Ibrahim, M., Bardes, A., Hoffman, J., Chellappa, R., Wilson, A. G., & Goldstein, T. (2023). Battle of the Backbones: A Large-Scale Comparison of Pretrained Models across Computer Vision Tasks. In 37th Conference on Neural Information Processing Systems (NeurIPS 2023).

Lee, Y., & Kang, P. (2022). AnoViT: Unsupervised Anomaly Detection and Localization with Vision Transformer-based Encoder-Decoder.

Liu, J., Xie, G., Wang, J., Li, S., Wang, C., Zheng, F., & Jin, Y. (2023). Deep Industrial Image Anomaly Detection: A Survey.

Mishra, P., Verk, R., Fornasier, D., Piciarelli, C., & Foresti, G. L. (2021). VT-ADL: A Vision Transformer Network for Image Anomaly Detection and Localization.

You, Z., Yang, K., Luo, W., Cui, L., Zheng, Y., & Le, X. (2022). ADTR: Anomaly Detection Transformer with Feature Reconstruction.

Zhu, J., Chen, X., He, K., LeCun, Y., & Liu, Z. (2025). Transformers without Normalization. *arXiv preprint arXiv:2503.10622*.

Mingxing Tan, Ruoming Pang, Quoc V. Le (2020). EfficientDet: Scalable and Efficient Object Detection